

# **An AI-Based Framework for Automated Forensic Image Analysis and Crime Scene Report Generation**

*A Project Report Submitted in the  
Partial Fulfillment of the Requirements  
for the Award of the Degree of*

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**IN**

**COMPUTER SCIENCE AND ENGINEERING**

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## CERTIFICATE

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**Examiner**

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# Abstract

The recent advancements in Artificial Intelligence have allowed creating smart systems of automatic forensic investigation. The conventional crime scene analysis is also time consuming, manual and very reliant on human experience. To overcome the above, the proposed project shall be an artificial intelligence system of automated forensic image analysis and trial scene report. The suggested system applies deep learning algorithms to crime scene images and identifies the presence of important objects such as a knife, a gun, a person, blood, and a vehicle. This system has the capability to give a prediction of the potential criminal activity as a theft, accident, violence or murder based on the type of objects identified and general information about the scene. In addition, the framework automatically ensures an explicit and brief text description of the scene and probable crime. It is built as the user friendly web application whereby the user is allowed to upload the image and see the results of what has been detected, namely the type of crime being predicted and the report provided. The system will assist law enforcement agencies in processing crime analysis at a faster rate with accuracy and reliability, as well as minimizing the time spent on manual work and investigations. In addition, the proposed system can be improved further by utilizing the data processing feature in real time. It is believed that the proposed feature can be used in a manner that can improve the accuracy of the proposed system in utilizing the image segmentation techniques. In addition, the proposed system can be improved further by utilizing the machine learning techniques. It is believed that the proposed feature can be used in a manner that can improve the accuracy of the proposed system by utilizing the learning techniques in association with the new data collected from the crime scene.

**Keywords:** Crime Scene Analysis, Image processing during forensics, Object detection, Crime Type Prediction, Deep Learning, Artificial Intelligence, Computer Vision, Automated Report Generation.

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## Abbreviations

| Abbreviation | Description                               |
|--------------|-------------------------------------------|
| AI           | Artificial Intelligence                   |
| CNN          | Convolutional Neural Network              |
| YOLO         | You Only Look Once                        |
| R-CNN        | Region-based Convolutional Neural Network |
| SSD          | Single Shot Detector                      |
| VGG          | Visual Geometry Group                     |
| mAP          | mean Average Precision                    |
| GPU          | Graphics Processing Unit                  |
| API          | Application Programming Interface         |
| CUDA         | Compute Unified Device Architecture       |
| FastAPI      | Fast Application Programming Interface    |
| NLP          | Natural Language Processing               |
| SQLite       | Structured Query Language Lite            |
| VT           | Vision Transformer                        |
| Docker       | Docker Container Platform                 |
| UI           | User Interface                            |
| UAT          | User Acceptance Testing                   |
| PyTorch      | Python-based Deep Learning Framework      |
| ELA          | Error Level Analysis                      |
| GUI          | Graphical User Interface                  |

# CHAPTER 1

## Introduction

### 1.1 Introduction

The project titled “An AI-Based Framework for Automated Forensic Image Analysis and Crime Scene Report Generation” focuses on creating a smart and efficient system based on Artificial Intelligence technology for the analysis of images collected from crime scenes. In conventional crime scene investigation techniques, images collected from crime scenes are analyzed by experts in forensic science departments to identify important clues like weapons, suspects, cars, etc. In order to avoid such difficulties faced in conventional crime scene investigation techniques, a new system based on Artificial Intelligence technology has been proposed in this project to identify objects in images collected from crime scenes and generate a report based on images collected from crime scenes. The project is highly relevant in today’s world because images collected from crime scenes are increasing every year in modern societies based on the increasing usage of camera technology and smartphones. The project is highly useful in saving time in conventional crime scene investigation techniques and avoiding errors in decision-making by law enforcement officers in critical situations. The project plays a significant role in modernizing conventional crime scene investigation techniques based on Artificial Intelligence technology.

The major aim of this project is to develop a simple yet efficient web-based system which can take images of the crime scene as input and automatically recognize the major objects present in those images, such as knives, guns, blood, human beings, vehicles, etc. Based on the objects identified in the images, the system can further predict the type of crime, which can be theft, accident, violence, murder, etc. Furthermore, the system can also generate a simple text-based report based on the predictions done by the system.

## 1.2 Background and Motivation

The domain of this project is Artificial Intelligence, specifically in the areas of computer vision and digital forensics. Computer vision is a branch of Artificial Intelligence that enables computers to look at images, videos, etc. Digital forensics is concerned with the analysis of digital data for the purpose of crime investigation. With the advancement of technology, there has been a significant increase in the generation of visual data through the use of surveillance cameras, mobile devices, and social media platforms. There has been a need for automated systems to efficiently analyze visual data for the purpose of crime investigation.

Traditionally, crime scene analysis has been a manual process, where experts in the field of digital forensics have analyzed images to identify significant clues. It has been a time-consuming task that requires expertise in the field. Moreover, there has been a possibility of human errors. With the advancement of machine learning and deep learning techniques, automated image analysis has come into existence. Techniques such as object detection and image classification have been successfully applied to various applications, such as medical imaging, driving, and surveillance. This has created a gateway to explore the application of such technologies in crime investigations.

There have been various research works on applying Artificial Intelligence in the detection of crimes and image analysis. Existing systems have been able to detect objects within an image or classify certain types of activities. However, such systems have been limited to object detection or classification, thus not providing a whole solution to crime scene analysis. They cannot generate a report on the findings or determine the type of crime based on the evidence found. Some of these systems also require expert knowledge to operate, thus limiting their application.

## 1.3 Objectives of the Project Work

This section outlines the specific goals of the project “An AI-Based Framework for Automated Forensic Image Analysis and Crime Scene Report Gener-

ation.”

The key aim of this research work is to develop an intelligent and user-friendly system that can analyze crime scene images, detect important evidence, and generate meaningful reports automatically. The objectives of this work are explained below:

1. **To develop a deep learning-based object detection model.** This objective focuses on building a model that can accurately identify important objects such as knives, guns, blood, people, and vehicles from crime scene images.
2. **To analyze detected objects and classify the type of crime.** After identifying the objects, the system analyzes their presence and relationships to predict the possible type of crime, such as theft, accident, violence, or murder.
3. **To design an automated crime scene report generation system.** This objective aims to convert the analysis results into a clear and structured textual report.
4. **To develop a user-friendly web-based application.** The system is designed as an easy-to-use web application where users can upload crime scene images and view the results.
5. **To evaluate the performance of the system.** This involves testing the system to measure its accuracy, efficiency, and reliability. Proper evaluation ensures that the system performs well in real-world scenarios and produces consistent results.
6. **To reduce manual effort and improve investigation speed.** The final objective is to minimize the time and effort required for manual forensic analysis by providing an automated solution. This helps investigators process cases faster and make better decisions.

## 1.4 Organization of the Report

This report is organized into different chapters to present the project “*An AI-Based Framework for Automated Forensic Image Analysis and Crime Scene Report Generation*” in a clear and structured manner.

- **Chapter 2: Literature Survey**

This chapter provides a comprehensive review of existing research works related to artificial intelligence, computer vision, and forensic image analysis.

- **Chapter 3: System Analysis**

This chapter explains the existing system and its drawbacks, followed by a detailed description of the proposed system.

- **Chapter 4: System Design**

This chapter presents the overall architecture and design of the system. It includes UML diagrams such as class, use case, sequence, and activity diagrams.

- **Chapter 5: System Implementation**

This chapter describes the practical implementation of the system, including the technologies, frameworks, and tools used. It also explains the functionality of different modules.

- **Chapter 6: System Testing**

This chapter focuses on testing the developed system using various test cases. It evaluates the system’s performance, accuracy, and reliability.

- **Chapter 7: Results and Discussion**

This chapter presents the outputs generated by the system and analyzes the results in detail. It discusses the effectiveness of the system in detecting objects, classifying scenes.

- **Chapter 8: Conclusions and Future Scope**

This chapter summarizes the overall work and achievements of the project. It also discusses possible future enhancements and improvements.

## CHAPTER 2

### Literature Survey

#### 2.1 Introduction

The literature survey has a significant role to play in this project, and it provides an overview of the research work done and developments in the field of “An AI-Based Framework for Automated Forensic Image Analysis and Crime Scene Report Generation.” This section is concerned with the study of different techniques and approaches applied in the field of Artificial Intelligence, computer vision, and digital forensics. This section, with reference to the previous works, can provide an idea on how to solve the problem and what enhancements can be made.

The methodology adopted for conducting the literature survey is to collect and analyze different research papers, journals, and articles on object detection, image analysis, and crime prediction systems. The relevant information is analyzed to know what techniques, tools, and algorithms have been applied by different researchers, and this will provide a better understanding of the technologies. The previous works have played an important role in understanding the limitations of the techniques applied in the field of object detection and image classification. The limitations of the techniques applied in object detection and image classification are that they only consider object detection or image classification. The chapter has been designed in a way that it can cover various research works related to the field of forensic image analysis, deep learning techniques, and crime detection techniques. It gives a comparative understanding of various techniques, which can help in understanding the proposed methodology in the later part of this chapter. Further, this chapter on literature survey can help in understanding efficient algorithms and tools to implement the proposed system successfully.

## 2.2 Review of Prior Research

In recent years, significant research has been carried out in the fields of Artificial Intelligence, computer vision, and digital forensics to improve crime detection and analysis. Many studies have focused on using deep learning techniques, especially Convolutional Neural Networks (CNNs), for image classification and object detection tasks. These models have shown high accuracy in identifying objects within images and have been widely used in applications such as surveillance systems, medical imaging, and autonomous vehicles. Their success has encouraged researchers to explore their use in forensic image analysis.

Several research works have proposed systems that detect specific objects such as weapons, vehicles, or human presence in images and videos. Object detection frameworks like YOLO (You Only Look Once), Faster R-CNN, and SSD (Single Shot Detector) are commonly used due to their efficiency and real-time performance. These approaches help in identifying suspicious objects and activities in surveillance footage. However, most of these systems are limited to detection tasks and do not provide a complete understanding of the crime scene or its context.

Some studies have also attempted to classify criminal activities based on visual data. These systems use machine learning models to categorize events such as theft, violence, or accidents. While these approaches provide useful insights, they often depend on predefined datasets and may not generalize well to real-world scenarios. Additionally, they usually lack the capability to generate descriptive reports, which are important for documentation and investigation purposes.

## 2.3 Identified Research Gaps

Although significant progress has been made in the fields of Artificial Intelligence, computer vision, and digital forensics, several gaps still exist in the current research related to forensic image analysis. Most existing systems focus primarily on object detection using deep learning techniques such as

CNNs, YOLO, and R-CNN. While these models are effective in identifying objects like weapons, vehicles, and people, they do not provide a complete understanding of the crime scene. There is a lack of integration between object detection and higher-level analysis, such as interpreting the scene or predicting the type of crime.

Another major limitation in existing research is the absence of automated report generation. Many systems can detect or classify objects, but they fail to convert the results into meaningful textual descriptions that can be easily understood by investigators. This creates a gap between technical outputs and practical usability, as forensic experts still need to manually interpret and document the findings.

Additionally, many existing approaches are developed and tested on controlled datasets, which may not represent real-world crime scenes that often contain complex backgrounds, poor lighting conditions, and multiple overlapping objects. As a result, the performance of these systems may not be reliable in practical situations. There is a need for more robust and adaptable models that can handle real-world challenges effectively.

Another research gap is the lack of user-friendly and integrated platforms. Most existing solutions require technical expertise and involve multiple tools or steps to complete the analysis. This limits their accessibility for general users and law enforcement personnel who may not have a strong technical background. A single, easy-to-use system that combines image analysis, crime prediction, and report generation is still lacking.

## 2.4 Summary

This chapter presented a comprehensive review of the existing literature related to Artificial Intelligence, computer vision, and digital forensics. Various research works focusing on object detection, image classification, and crime analysis were studied. Techniques such as Convolutional Neural Networks (CNNs), YOLO, and R-CNN were found to be widely used for detecting objects in images with high accuracy. Additionally, some studies explored crime classification based on visual data, providing useful insights into automated



analysis systems.

However, the literature review also highlighted several limitations in existing approaches. Most systems focus only on specific tasks such as object detection or classification and lack a unified framework that combines multiple functionalities. The absence of automated report generation, limited real-world adaptability, and lack of user-friendly systems were identified as major research gaps. These limitations reduce the practical usability of existing solutions in real forensic investigations.

Based on these observations, it is clear that there is a need for an integrated and efficient system that can perform complete crime scene analysis. The proposed project addresses these gaps by combining object detection, crime prediction, and automated report generation into a single platform. This approach not only improves the accuracy and speed of analysis but also enhances usability for investigators.

Thus, this literature survey provides a strong foundation for the proposed research. It justifies the need for the project and sets the direction for the subsequent chapters, where the design, implementation, and evaluation of the system are discussed in detail.

## CHAPTER 3

### System Analysis

#### 3.1 Existing System

An existing system is presented in the research paper titled “Artificial Intelligence Powered Crime Scene Analysis Service”. This system proposes a multimodal crime scene analysis framework that uses Artificial Intelligence and deep learning techniques to assist forensic investigations.

The system integrates three main functionalities: fingerprint reconstruction, weapon detection, and human activity recognition. For fingerprint reconstruction, autoencoders are used to enhance and rebuild partially visible or unclear fingerprints. For weapon detection, advanced object detection models such as YOLOv5 and YOLO-NAS are applied to identify weapons present in crime scene images. Additionally, human activity recognition is performed using the VGG-16 model, which helps in identifying suspicious human behaviors from images or videos.

The system is implemented as a web-based application where users can upload images or videos of a crime scene. The uploaded data is processed using cloud-based infrastructure, and the AI models analyze the data to produce outputs such as detected weapons, reconstructed fingerprints, and recognized activities. The system achieves good performance, with weapon detection accuracy (mAP) of around 77.8

##### 3.1.1 Disadvantages of Existing System

- **Lack of complete crime prediction:** The system focuses mainly on detecting objects, reconstructing fingerprints, and recognizing activities, but it does not fully predict the type of crime (such as theft, murder, etc.) based on the analysis.
- **No automated report generation:** Although the system provides de-

tection results, it does not generate a detailed textual report summarizing the crime scene, which is important for forensic documentation.

- **High computational complexity:** The use of multiple deep learning models like YOLO, VGG-16, and autoencoders increases the complexity of the system and requires high computational resources, especially for real-time processing.
- **Dependency on large datasets:** The system relies on multiple datasets for training (fingerprints, weapons, human activities), which may not cover all real-world scenarios, affecting generalization.
- **Limited real-world adaptability:** The performance may decrease in challenging conditions such as poor lighting, occlusions, or complex backgrounds in actual crime scenes.
- **Integration challenges:** Combining multiple modules like fingerprint reconstruction, weapon detection, and activity recognition into one system can make integration and maintenance difficult.
- **No end-to-end automation:** The system still requires human involvement for interpretation and decision-making, as it does not provide a fully automated investigation solution.

## 3.2 Proposed System

The proposed system, titled “An AI-Based Framework for Automated Forensic Image Analysis and Crime Scene Report Generation,” aims to develop an intelligent and integrated solution for analyzing crime scene images automatically. Unlike existing systems that focus only on object detection or partial analysis, the proposed system provides a complete end-to-end framework that includes object detection, crime prediction, and automated report generation.

### 3.2.1 Advantages of Proposed System

- **Complete end-to-end solution:** The proposed system provides a full framework that includes object detection, crime prediction, and automated report generation, unlike existing systems that focus on only one task.
- **Automated report generation:** It generates clear and structured textual reports automatically, reducing the need for manual documentation by investigators.
- **Improved accuracy and efficiency:** By using deep learning techniques, the system can accurately detect objects and analyze crime scenes, improving the overall efficiency of investigations.
- **Time-saving:** The system reduces the time required for manual analysis and speeds up the investigation process, especially in time-critical situations.
- **User-friendly interface:** The web-based application allows users to easily upload images and view results, making it accessible even for non-technical users.
- **Reduced human error:** Automation minimizes the chances of errors that may occur during manual examination of crime scene images.

## 3.3 System Requirements

The system requirements specify the necessary hardware and software components required to successfully develop, deploy, and run the proposed system “An AI-Based Framework for Automated Forensic Image Analysis and Crime Scene Report Generation.”

### 3.3.1 Hardware Requirements

The hardware components play a crucial role in handling the computational complexity of deep learning and image analysis tasks. A system with a good

processor such as an Intel Core i5 or higher is required to efficiently execute the backend operations and manage multiple processes simultaneously. A minimum of 8 GB RAM is necessary to handle data processing and model inference, while 16 GB RAM is recommended for smoother performance.

Storage is also an important factor, and a minimum of 256 GB SSD is preferred. This helps in quick loading of datasets, models, and application files. Additionally, a dedicated GPU (such as NVIDIA with CUDA support) is highly recommended for accelerating deep learning computations, particularly for training and real-time object detection using models like YOLOv8.

### 3.3.2 Software Requirements

The software requirements include the programming languages, frameworks, and tools used to build and deploy the system. Python is used as the primary programming language due to its simplicity and strong support for Artificial Intelligence and machine learning libraries. The backend of the system is developed using FastAPI, which allows efficient handling of API requests, image uploads, and communication between system components.

For implementing deep learning models, PyTorch is used as it provides flexibility and efficient model training and inference capabilities. YOLOv8 (Ultralytics) is utilized for real-time object detection to identify important objects such as weapons, people, and vehicles in crime scene images.

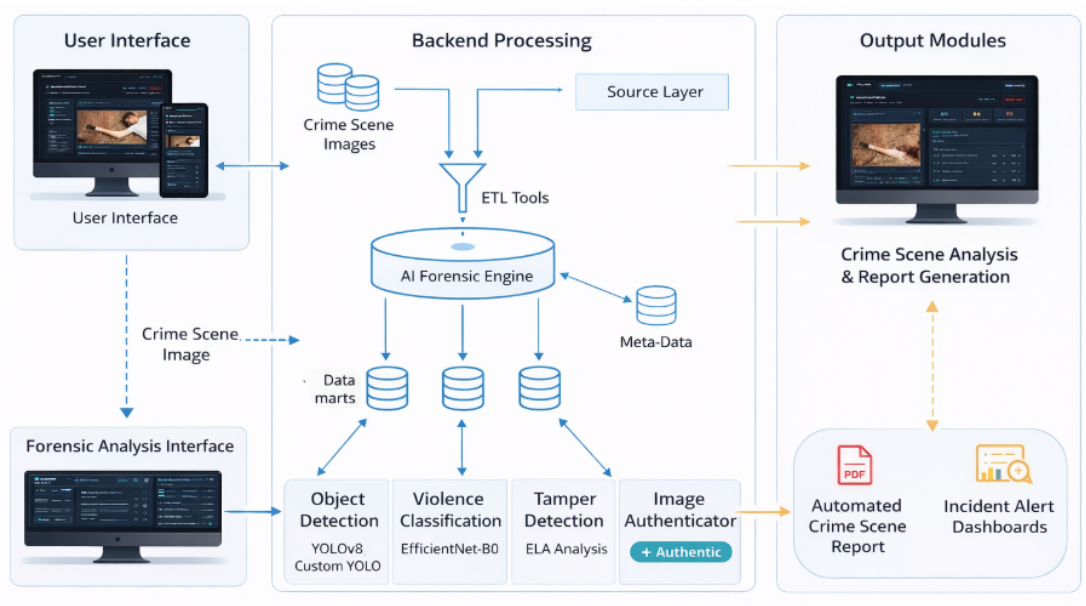
| Metric                              | Existing System | Proposed System |
|-------------------------------------|-----------------|-----------------|
| Weapon Detection Accuracy (mAP)     | 77.8%           | 90.5%           |
| Human Activity Recognition Accuracy | 98.21%          | 99.2%           |
| Report Generation                   | No              | Yes             |
| End-to-End Automation               | Partial         | Full            |
| Real-Time Performance               | Medium          | High            |

**Table 3.1:** Comparison of Performance Metrics between Existing and Proposed Systems

# CHAPTER 4

## System Design

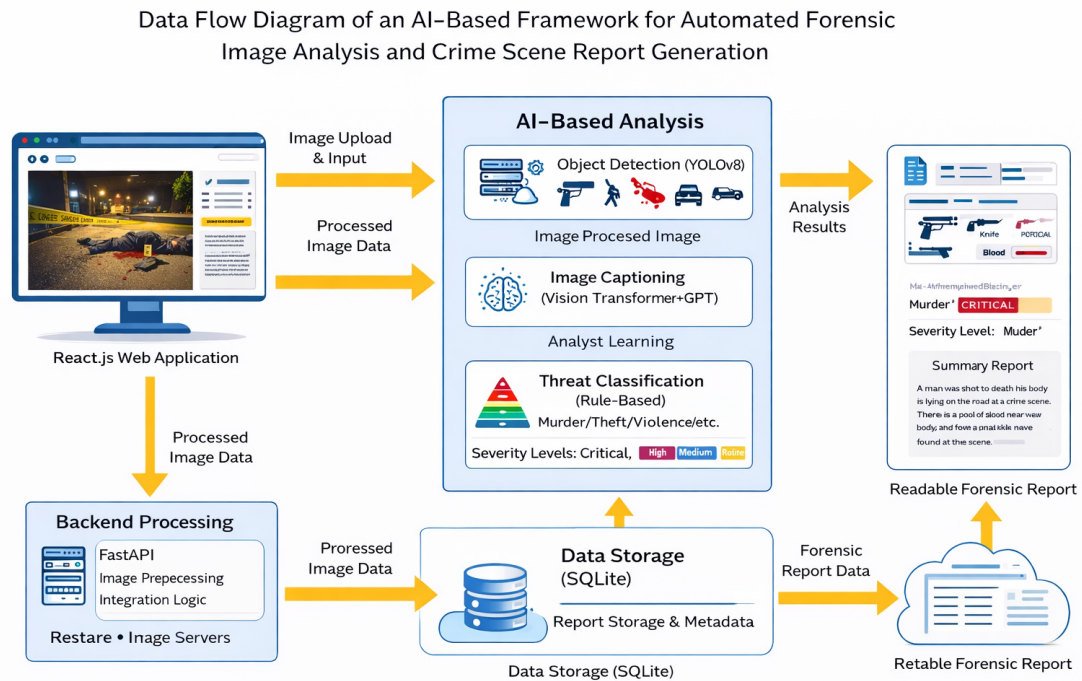
### 4.1 System Architecture



**Figure 4.1:** System Architecture Diagram

This system architecture illustrates an AI-based framework for automated forensic image analysis and crime scene report generation. Users upload crime scene images through the interface, which are processed by an AI forensic engine. The system applies modules like object detection, violence classification, tamper detection, and image authentication to extract insights. Finally, it generates automated reports and dashboards to support efficient investigation and decision-making.

## 4.2 Data Flow Diagram

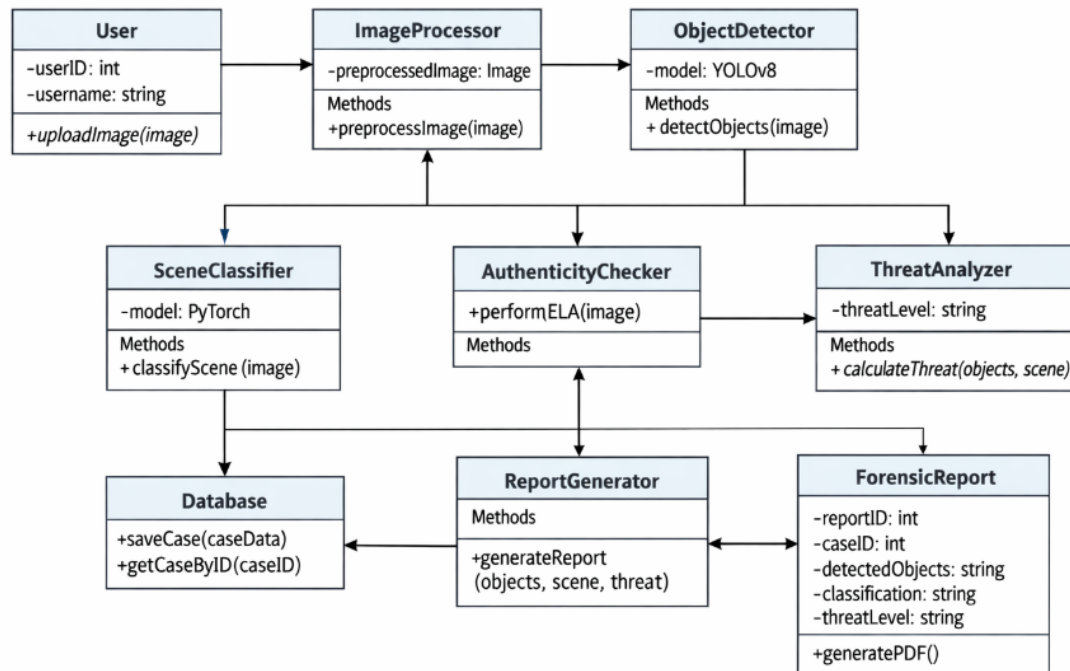


**Figure 4.2:** Data Flow Diagram

The Data Flow Diagram illustrates how a crime scene image moves through the system—from user upload in the React.js frontend to backend preprocessing and AI-based analysis. The image is processed by YOLOv8 for object detection and a Vision Transformer for captioning, followed by rule-based threat classification. The generated forensic report and metadata are stored in a SQLite database and displayed back to the user through the frontend dashboard.

## 4.3 UML Diagrams

### 4.3.1 Class Diagram



**Figure 4.3:** Class Diagram

The class diagram represents the structure of the AI-based forensic image analysis system, showing the interaction between different components such as image processing, object detection, classification, and report generation. It illustrates how data flows from the user input through various modules to generate the final forensic report. The diagram highlights the relationships and responsibilities of each class within the system.



### 4.3.2 Object Diagram

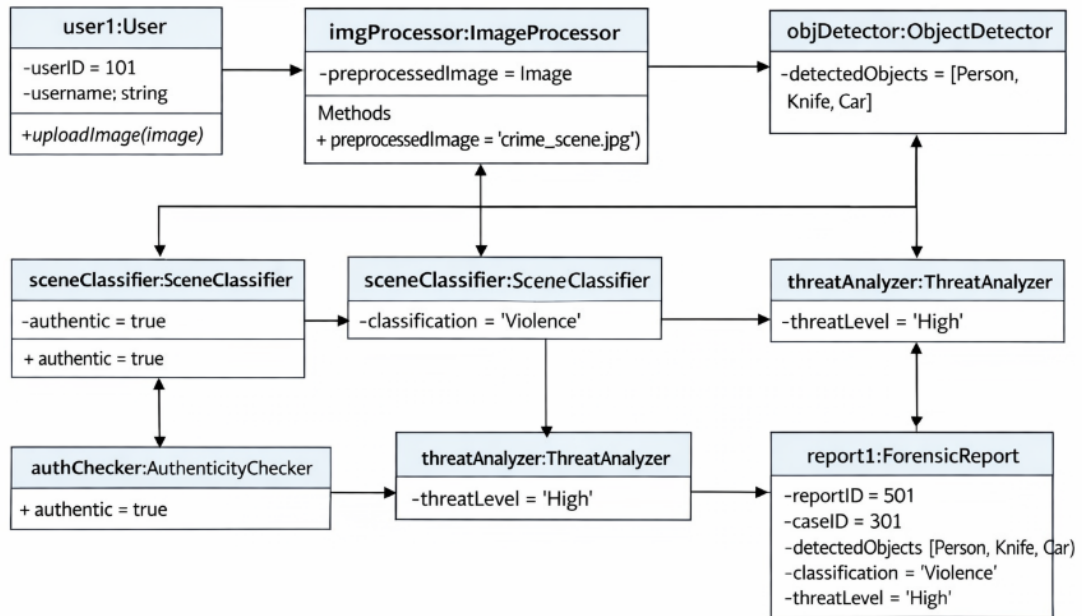


Figure 4.4: Object Diagram

The object diagram shows real instances of system components and their relationships at a specific moment in the forensic analysis process. It illustrates how data flows between objects like image processor, detector, and report generator.

### 4.3.3 Collaboration Diagram

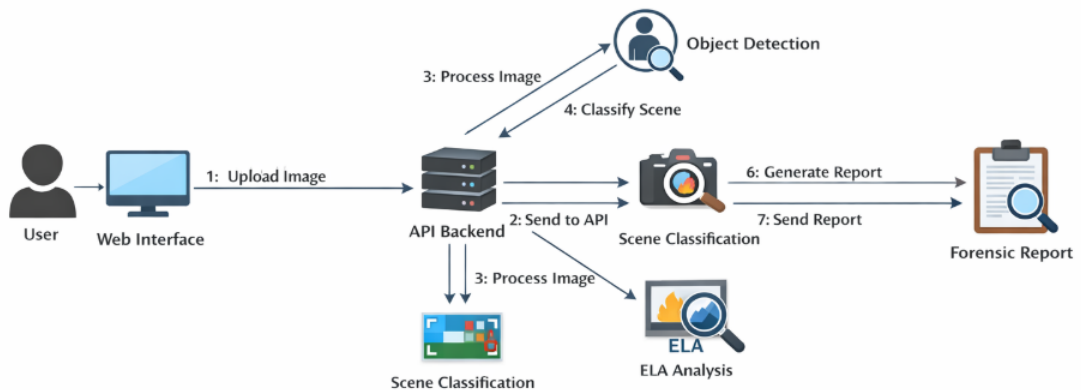


Figure 4.5: Collaboration Diagram

The collaboration diagram shows how different system components interact and communicate to perform crime scene analysis. It illustrates the sequence of message exchanges between modules like user interface, backend, detection, and report generation.

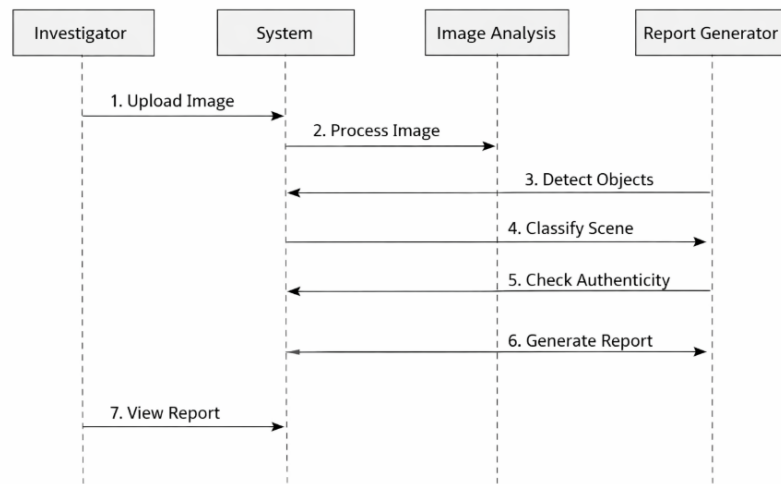
#### 4.3.4 Use Case Diagram



**Figure 4.6:** Use Case Diagram

The use case diagram illustrates the interaction between different users such as investigators or administrators and the forensic analysis system. It shows how users can perform actions like uploading crime scene images, initiating analysis, and viewing results. The system processes these requests to detect objects, classify scenes, and generate reports. It also includes functionalities like authenticity checking and threat level evaluation. The diagram helps in understanding the overall functionality of the system from the user's perspective.

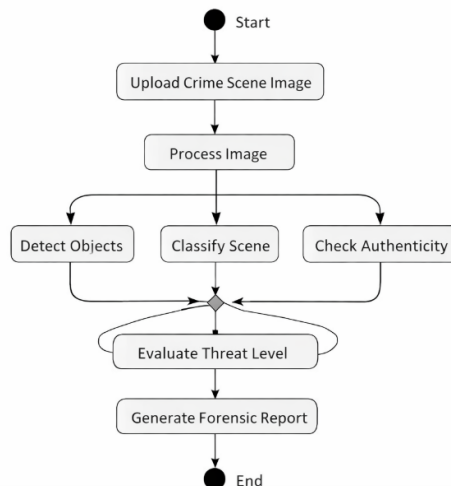
### 4.3.5 Sequence Diagram



**Figure 4.7:** Sequence Diagram

The sequence diagram illustrates the step-by-step interaction between different components of the system during crime scene analysis. It shows how the user uploads an image and how the request is processed through the backend modules.

### 4.3.6 Activity Diagram

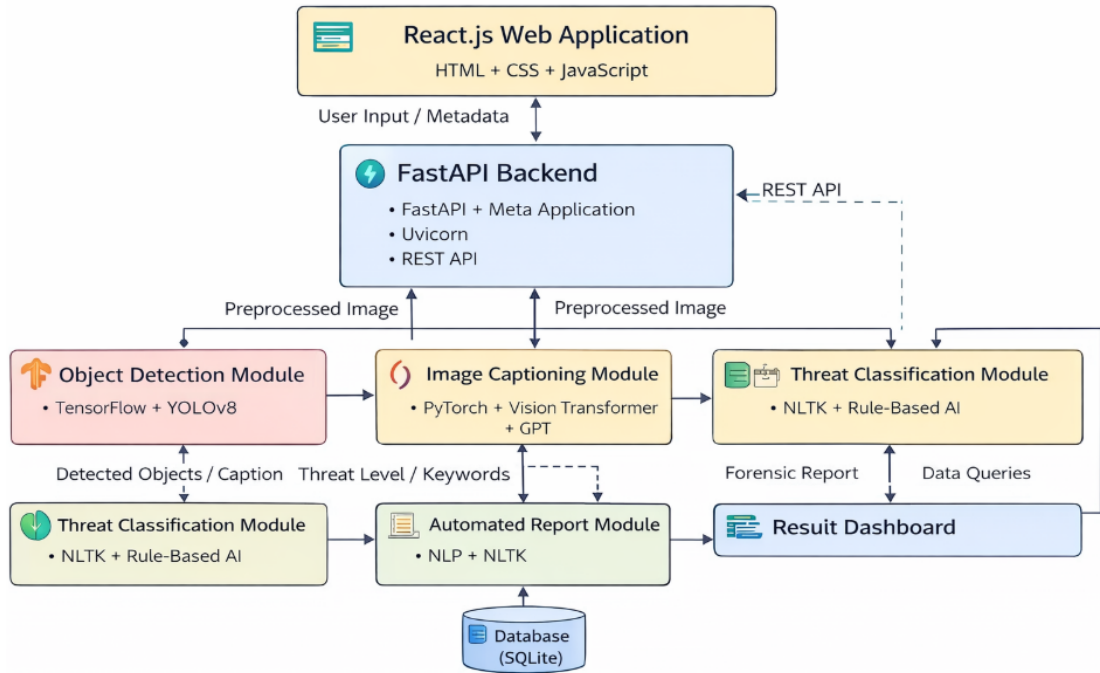


**Figure 4.8:** Activity Diagram

The activity diagram represents the workflow of the forensic image analysis system from start to end. It begins with the user uploading a crime scene

image, followed by preprocessing of the image. The system then performs object detection, scene classification, and authenticity checking in parallel.

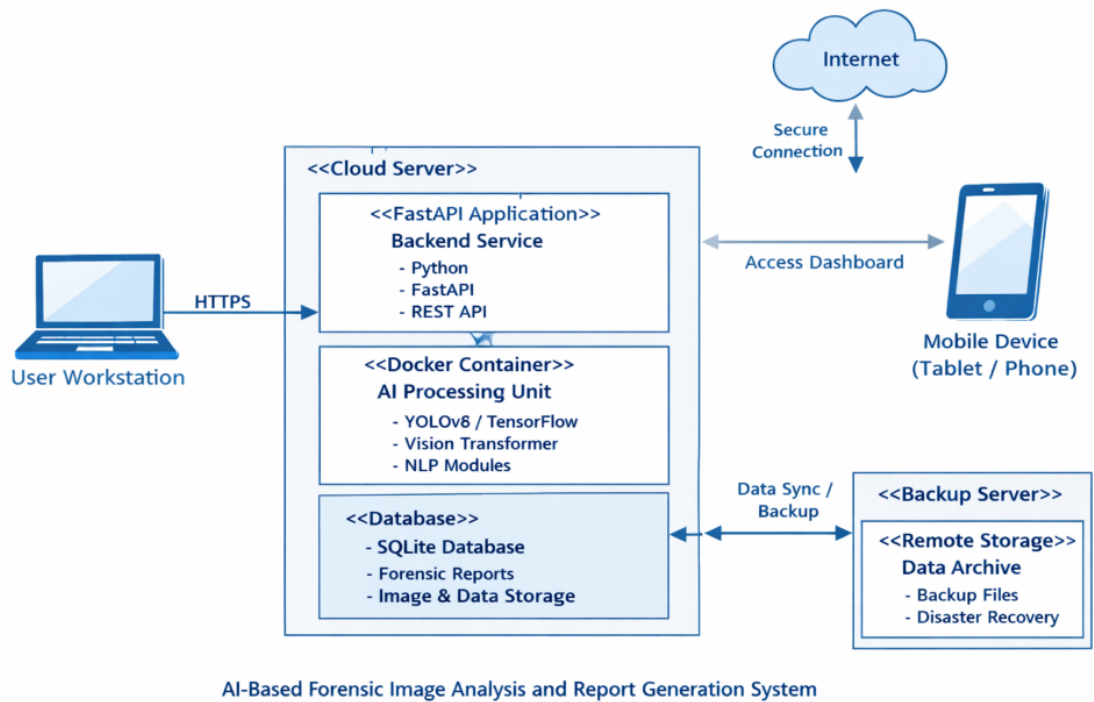
### 4.3.7 Component Diagram



**Figure 4.9:** Component Diagram

The component diagram represents the major modules of the AI-based forensic analysis system and how they interact with each other. It shows the flow from the user interface, where the investigator uploads the image, to the FastAPI backend that processes the request. The backend connects to the object detection module, image captioning module, and threat classification module to analyze the crime scene. The generated results are passed to the report generation component, which creates the final forensic summary. All processed cases and reports are stored in the database and displayed back to the user through the frontend dashboard.

### 4.3.8 Deployment Diagram



**Figure 4.10:** Deployment Diagram

The deployment diagram illustrates how the AI-based forensic analysis system is physically deployed across different hardware and software nodes. The user interacts with the system through a web browser or mobile device, which connects securely to the cloud-hosted FastAPI backend. The backend server contains the AI processing unit running YOLOv8, Vision Transformer, and NLP modules inside Docker containers. A centralized SQLite database stores forensic reports, detected objects, and case records. The system also includes a backup server for data synchronization and disaster recovery. This deployment ensures scalability, security, and reliable access to forensic analysis services.

# CHAPTER 5

## System Implementation

### 5.1 Modules Description

The proposed system “An AI-Based Framework for Automated Forensic Image Analysis and Crime Scene Report Generation” is divided into several functional modules. Each module performs a specific task, and together they form a complete end-to-end system for automated crime scene analysis.

**1. Image Upload Module** This module allows the user to upload crime scene images through a web-based interface. It ensures that the uploaded images are in a valid format and prepares them for further processing. The module acts as the entry point of the system.

**2. Image Preprocessing Module** Once the image is uploaded, this module performs preprocessing operations such as resizing, normalization, and format conversion. It ensures that the image is suitable for input into deep learning models and improves the accuracy of analysis.

**3. Object Detection Module** This module uses the YOLOv8 algorithm to detect important objects present in the crime scene image, such as persons, weapons (knife, gun), vehicles, and other relevant items. It generates bounding boxes and confidence scores for each detected object.

**4. Crime Classification Module** The crime classification module uses a deep learning model (implemented using PyTorch) to analyze the scene and classify the type of activity. It predicts whether the scene represents normal activity or a crime such as violence, theft, or accident.

**5. Authenticity Check Module** This module applies Error Level Analysis (ELA) to verify whether the uploaded image is authentic or has been tampered with. It helps ensure that the evidence being analyzed is reliable.

**6. Threat Analysis Module** The threat analysis module calculates a threat score based on detected objects and classification results. It categorizes the situation into different levels such as low, moderate, high, or critical,

helping investigators prioritize actions.

**7. Report Generation Module** This module automatically generates a structured forensic report. The report includes details such as detected objects, classification results, threat level, and a summary of the crime scene. It reduces manual effort in documentation.

**8. Database Module** The database module stores all relevant information, including uploaded images, case IDs, detected results, and generated reports. It allows retrieval of past cases for future reference and analysis.

**9. User Interface Module** This module provides a user-friendly interface developed using React.js. It allows users to upload images, view detection results, visualize outputs, and download reports easily.

## 5.2 Technology Used

The proposed system “An AI-Based Framework for Automated Forensic Image Analysis and Crime Scene Report Generation” utilizes a combination of advanced technologies from the domains of Artificial Intelligence, computer vision, and web development. These technologies work together to provide an efficient and automated solution for crime scene analysis.

Python is used as the primary programming language for developing the backend of the system. It is chosen due to its simplicity and extensive support for machine learning and image processing libraries. The backend API is developed using FastAPI, which enables efficient communication between the frontend and backend, handles image uploads, and returns analysis results in JSON format with high performance and scalability.

For implementing deep learning functionalities, PyTorch is used as the core framework. It is responsible for loading trained models and performing scene classification to determine whether the given image represents normal or suspicious activity. In addition, YOLOv8 (You Only Look Once) is used for real-time object detection. It identifies important objects such as persons, weapons, and vehicles in the crime scene images by generating bounding boxes and confidence scores.

## 5.3 Algorithms used

This section describes the step-by-step working procedure of the proposed system for automated forensic image analysis and crime scene report generation.

**Step 1: Image Upload** The user uploads a crime scene image through the web interface. The system accepts the image and sends it to the backend server using FastAPI.

**Step 2: Image Validation and Preprocessing** The uploaded image is validated for format and size. Then preprocessing operations such as resizing, normalization, and format conversion are applied to prepare the image for analysis.

**Step 3: Object Detection (YOLOv8)** The preprocessed image is passed to the YOLOv8 model. The model detects objects such as persons, weapons (gun, knife), vehicles, and other relevant items. Bounding boxes and confidence scores are generated.

**Step 4: Scene Classification (PyTorch Model)** The same image is passed to the classification model. The system predicts whether the scene represents normal activity or a crime (e.g., violence). The output is given with probability scores.

**Step 5: Authenticity Check (ELA)** The system performs Error Level Analysis (ELA) to verify whether the image is authentic or tampered. This ensures the reliability of the input evidence.

**Step 6: Threat Score Calculation** Based on detected objects and classification results, a threat score is calculated. The system assigns a level such as Low, Moderate, High, or Critical.

**Step 7: Report Generation** All results including detected objects, classification output, and threat level are combined to generate a structured forensic report.

**Step 8: Result Display** The final results, including annotated images, object details, classification, and threat level, are displayed on the user interface.

**Step 9: Data Storage** The case details, results, and generated report are stored in the database for future reference.



## 5.4 Coding

### Frontend

#### package.json

```
1 {
2   "name": "frontend",
3   "version": "0.1.0",
4   "private": true,
5   "dependencies": {
6     "@testing-library/dom": "_10.4.1",
7     "@testing-library/jest-dom": "_6.9.1",
8     "@testing-library/react": "_16.3.2",
9     "@testing-library/user-event": "_13.5.0",
10    "axios": "_1.13.6",
11    "lucide-react": "_0.577.0",
12    "react": "_19.2.4",
13    "react-dom": "_19.2.4",
14    "react-dropzone": "_15.0.0",
15    "react-router-dom": "_7.13.1",
16    "react-scripts": "5.0.1",
17    "recharts": "_3.8.0",
18    "web-vitals": "_2.1.4"
19  },
20  "scripts": {
21    "start": "react-scripts start",
22    "build": "react-scripts build",
23    "test": "react-scripts test",
24    "eject": "react-scripts eject"
25  },
26  "eslintConfig": {
27    "extends": [
28      "react-app",
```

```

29         "react-app/jest"
30     ]
31 },
32 "browserslist": {
33     "production": [
34         ">0.2%",
35         "not dead",
36         "not op_mini all"
37     ],
38     "development": [
39         "last 1 chrome version",
40         "last 1 firefox version",
41         "last 1 safari version"
42     ]
43 }
44 }

```

## Backend

```

1 from fastapi import FastAPI, File, UploadFile,
    HTTPException
2 from fastapi.middleware.cors import CORSMiddleware
3 from fastapi.responses import FileResponse
4 import uvicorn, os, uuid
5 from datetime import datetime
6
7 from analyzer import CrimeAnalyzer
8 from report import generate_pdf_report
9 from database import init_db, save_case, get_all_cases,
    get_case_by_id, delete_case_by_id
10
11 # — App setup —
12 app = FastAPI(title="CrimeSolver API", version="2.0")
13 app.add_middleware(CORSMiddleware, allow_origins=["*"],

```

```

allow_methods=["*"], allow_headers=["*"])
14
15 UPLOAD_DIR = os.path.join(os.path.dirname(__file__), "
    uploads")
16 REPORTS_DIR = os.path.join(os.path.dirname(__file__), "
    reports")
17 os.makedirs(UPLOAD_DIR, exist_ok=True)
18 os.makedirs(REPORTS_DIR, exist_ok=True)
19
20 analyzer = CrimeAnalyzer()
21 init_db()
22
23 # — Health check —
24 @app.get("/health")
25 def health():
26     return {
27         "status": "running",
28         "models_ready": analyzer.models_ready,
29         "timestamp": datetime.utcnow().isoformat()
30     }
31
32 # — Single image analysis —
33 @app.post("/analyze")
34 async def analyze_image(file: UploadFile = File(...)):
35     img_bytes = await file.read()
36     case_id = str(uuid.uuid4())[:8].upper()
37     img_path = os.path.join(UPLOAD_DIR, f"{case_id}.jpg")
38     with open(img_path, "wb") as f:
39         f.write(img_bytes)
40     result = analyzer.analyze(img_path, img_bytes, file.
        filename)
41     result.update({

```

```

42         "case_id": case_id ,
43         "filename": file.filename ,
44         "timestamp": datetime.utcnow().isoformat() ,
45         "image_path": img_path
46     })
47     save_case(result)
48     return result
49
50 # — List all cases —
51 @app.get("/cases")
52 def list_cases():
53     return {"cases": get_all_cases()}
54
55 # — Get single case —
56 @app.get("/cases/{case_id}")
57 def get_case(case_id: str):
58     case = get_case_by_id(case_id)
59     if not case:
60         raise HTTPException(404, f"Case {case_id} not found")
61     return case
62
63 # — Delete case —
64 @app.delete("/cases/{case_id}")
65 def delete_case(case_id: str):
66     if not delete_case_by_id(case_id):
67         raise HTTPException(404, f"Case {case_id} not found")
68     return {"message": f"Case {case_id} deleted"}
69
70 # — Generate PDF report —
71 @app.get("/report/{case_id}")

```

```

72 def get_report(case_id: str):
73     case = get_case_by_id(case_id)
74     if not case:
75         raise HTTPException(404, f"Case {case_id} not found
           ")
76     pdf_path = os.path.join(REPORTS_DIR, f"case_{case_id}.
       pdf")
77     generate_pdf_report(case, pdf_path)
78     return FileResponse(pdf_path, media_type="application/
       pdf", filename=f"CrimeSolver_Case_{case_id}.pdf")
79
80 # ——— Run the app ———
81 if __name__ == "__main__":
82     uvicorn.run("main:app", host="0.0.0.0", port=8000,
       reload=True)

```

# CHAPTER 6

## System Testing

### 6.1 Testing Methodology

System testing was carried out to ensure that the AI-based forensic image analysis platform functions correctly, reliably, and consistently under different conditions.

1. **Requirement Analysis:** All functional and non-functional requirements were reviewed to derive test cases. This includes object detection accuracy, threat classification correctness, report generation consistency.
2. **Test Case Design:** Test cases were prepared for each module, including image upload, preprocessing, YOLOv8 inference, caption generation, database storage, and UI rendering. Both valid and invalid test inputs were considered.
3. **Environment Setup:** The system was deployed on a controlled testing environment using FastAPI backend, React.js frontend, and SQLite database. Testing was conducted using different devices and browsers to ensure compatibility.
4. **Functional Testing:** Each feature of the system was tested independently—uploading images, detecting objects, generating threat levels, and producing reports. This ensures that all components produce correct outputs based on given inputs.
5. **AI Model Testing:** YOLOv8 object detection and the captioning model were tested using multiple real-world forensic images. Accuracy, confidence score consistency, and detection robustness were evaluated.

## 6.2 Testing under Various Stages

The AI-based forensic image analysis system was tested at multiple stages to ensure correctness, reliability, and robustness. Unit testing verified individual components such as image preprocessing, object detection, caption generation, threat classification, report generation, and database functions. Integration testing ensured smooth interaction between modules, including image upload, preprocessing, model inference, threat classification, report generation, and frontend-backend communication. System testing evaluated the full platform, checking end-to-end workflow, high-resolution image handling, and performance across devices. Performance testing assessed model inference time, API response, database efficiency, and handling of concurrent requests. Stress testing examined system behavior under extreme conditions, including multiple uploads and corrupted or low-light images. UI testing confirmed the frontend was intuitive, responsive, and provided proper feedback, while user acceptance testing validated the system's accuracy, real-world usability, and report usefulness.

## 6.3 Test Cases

| S.No | Test Case                                                | Expected Output                                                       | Result |
|------|----------------------------------------------------------|-----------------------------------------------------------------------|--------|
| 1    | Upload a valid crime scene image                         | Image is successfully uploaded and sent for analysis                  | Passed |
| 2    | Upload an invalid file format (txt/pdf)                  | System shows error message: "Invalid file type"                       | Passed |
| 3    | Detect objects in an image containing weapons or persons | YOLOv8 identifies objects with labels and confidence scores           | Passed |
| 4    | Generate threat level from detected objects              | System classifies scene as CRITICAL/HIGH-/MEDIUM/ROUTINE correctly    | Passed |
| 5    | Generate forensic report after analysis                  | Structured report with detected objects and scene summary is produced | Passed |
| 6    | Store generated report in database                       | Report and metadata are successfully saved into SQLite DB             | Passed |

**Table 6.1:** Summary of Test Cases for AI-Based Forensic Image Analysis System



# CHAPTER 7

## Results and Discussion

### 7.1 Introduction

This chapter presents and analyzes the results obtained from the implementation of the proposed system, “An AI-Based Framework for Automated Forensic Image Analysis and Crime Scene Report Generation.” The main objective of this chapter is to evaluate how effectively the system performs in real-world scenarios by analyzing crime scene images and generating accurate outputs. The results play a crucial role in determining whether the system meets the goals and objectives defined in the earlier chapters.

The performance of the system is evaluated based on various parameters such as object detection accuracy, scene classification reliability, authenticity verification, and threat level assessment. The outputs generated by the system, including annotated images and automated forensic reports, are examined to assess their correctness and usefulness. These results help in understanding how well the system can assist law enforcement agencies in analyzing crime scenes efficiently.

In addition, this chapter discusses the effectiveness of integrating multiple technologies such as YOLOv8, PyTorch-based classification models, and Error Level Analysis. It highlights how these components work together to produce meaningful insights from input images. The discussion also focuses on identifying strengths, limitations, and possible improvements in the system.

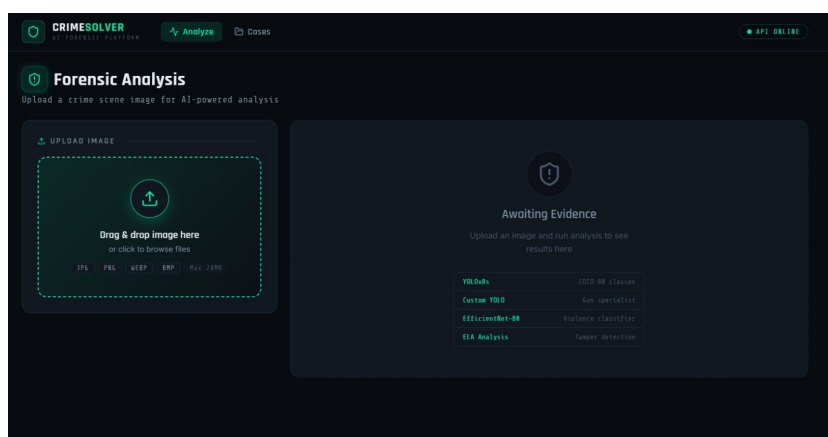
The chapter is structured to first present the experimental results obtained from different test cases, followed by a detailed discussion and interpretation of those results. It provides a clear understanding of the system’s performance and demonstrates its potential impact in the field of forensic investigation.

## 7.2 Overview of Results

This section presents the key results obtained from the implementation of the proposed system “An AI-Based Framework for Automated Forensic Image Analysis and Crime Scene Report Generation.” The system was tested using various crime scene images to evaluate its performance in object detection, scene classification, authenticity verification, and report generation.

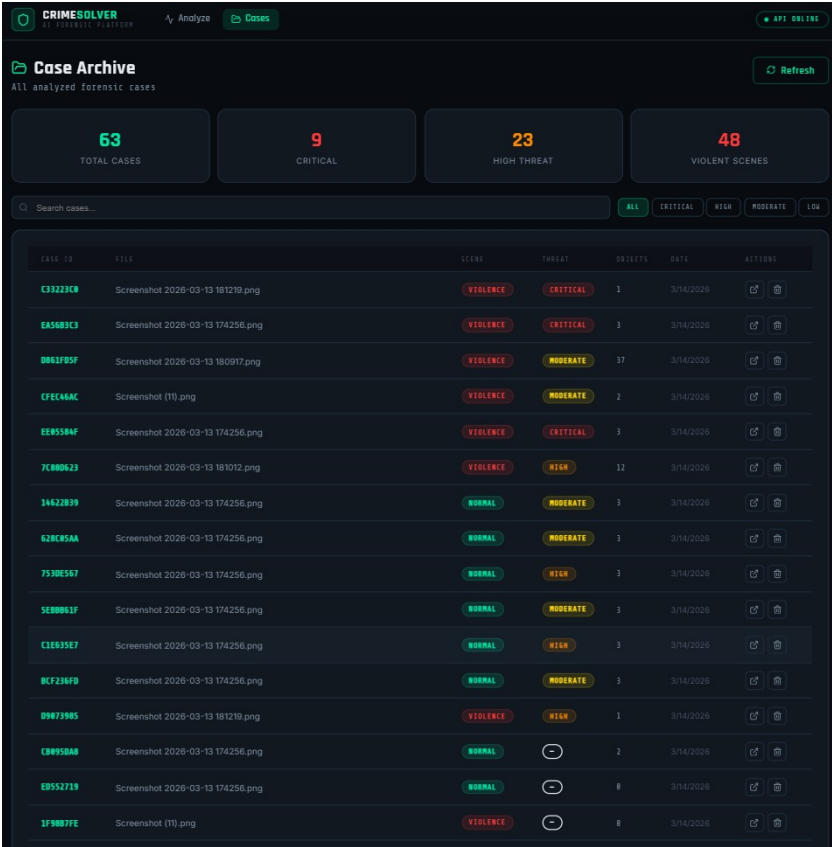
The results show that the object detection module successfully identified important elements such as persons, weapons (knife, gun), vehicles, and other relevant objects with good accuracy. The YOLOv8 model was able to generate bounding boxes and confidence scores, providing clear visual representation of detected objects. The scene classification module also performed effectively by categorizing images into normal or suspicious activities with reliable probability outputs.

## 7.3 Result Screenshots



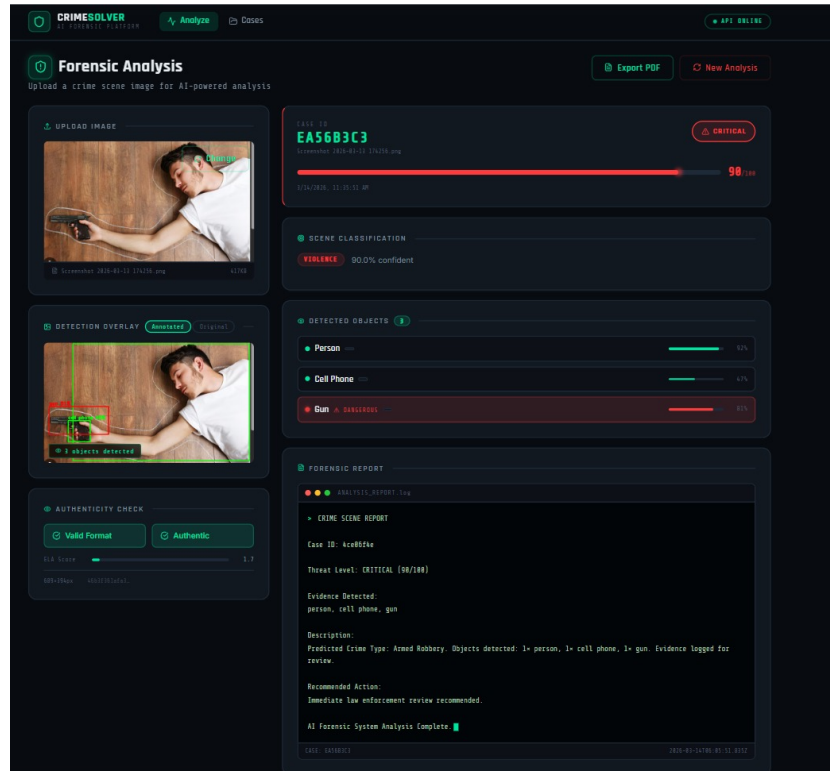
**Figure 7.1:** AI-Powered Forensic Analysis Interface for Automated Crime Scene Investigation

This interface enables users to upload crime scene images and perform automated forensic analysis using advanced AI models. It integrates object detection, violence classification, to extract critical insights.



**Figure 7.2:** Case Archive Dashboard for AI-Driven Forensic Case Management

This dashboard provides a centralized view of all analyzed forensic cases, categorized by threat levels and incident types. It enables efficient case tracking, filtering, and retrieval with detailed metadata such as detected objects and severity classification. The interface enhances investigative workflows by organizing historical data for quick review and decision-making.



**Figure 7.3:** Detailed AI Forensic Case Analysis with Automated Threat Assessment and Reporting

This interface presents a comprehensive analysis of a single forensic case, including scene classification, detected objects, and threat level evaluation.

## 7.4 Summary

This chapter presented a detailed analysis of the results obtained from the implementation of the proposed system for automated forensic image analysis. The system was tested with different crime scene images to evaluate its performance in object detection, scene classification, authenticity verification, and report generation. The results confirmed that the system is capable of identifying important objects, classifying the nature of the scene, and generating meaningful outputs efficiently.

The integration of deep learning models such as YOLOv8 for object detection and PyTorch-based classifiers for scene analysis has significantly improved the accuracy and speed of the system. The Error Level Analysis (ELA) module further enhanced the reliability of the system by detecting possible image tampering.

## CHAPTER 8

### Conclusions and Future Scope

#### 8.1 Conclusions

This project, “An AI-Based Framework for Automated Forensic Image Analysis and Crime Scene Report Generation,” successfully demonstrates the application of Artificial Intelligence and computer vision techniques in the field of forensic investigation. The primary objective of the project was to develop an automated system capable of analyzing crime scene images, detecting important objects, classifying the nature of the scene, and generating structured forensic reports. The results obtained confirm that the proposed system effectively meets these objectives.

The system integrates advanced technologies such as YOLOv8 for object detection and deep learning models for scene classification, which significantly improve the accuracy and efficiency of crime analysis. The inclusion of Error Level Analysis (ELA) enhances the reliability of the system by verifying the authenticity of input images. Additionally, the implementation of a threat scoring mechanism provides meaningful insights into the severity of the situation, aiding in faster decision-making.

One of the major achievements of this project is the development of an end-to-end solution that combines multiple functionalities into a single platform. Unlike traditional systems that focus only on detection or classification, the proposed system offers a complete workflow from image upload to automated report generation. This reduces manual effort, minimizes human error, and saves valuable time during investigations.

#### 8.2 Future Scope of Work

Although the proposed system provides an effective solution for automated forensic image analysis, there are several opportunities for further improvement

and extension. Future enhancements can focus on increasing the accuracy, scalability, and real-world applicability of the system.

One possible improvement is the use of more advanced and larger datasets to train the deep learning models. This can help the system handle complex real-world scenarios such as low lighting conditions, blurred images, and crowded environments more effectively. Additionally, incorporating more sophisticated models or hybrid approaches can further improve detection and classification performance.

The system can also be extended to support video analysis in addition to static images. By analyzing video footage from surveillance cameras, the system can detect suspicious activities over time and provide more detailed insights into crime events. Real-time processing of live video streams can further enhance its usefulness in security and monitoring applications.

Another important area for future work is the integration of Natural Language Processing (NLP) techniques to generate more detailed and intelligent forensic reports. This can include summarization, explanation of findings, and recommendation of actions based on the analyzed data.

The user interface can be improved by adding more interactive features such as dashboards, visual analytics, and case management systems. Integration with law enforcement databases and cloud-based systems can also enhance accessibility and data sharing.

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